Laplace Transforms (L.T.)

Pierre Simmon M. De. → L.T. (1749 – 1827)

Uses:- ① particular solution is obtained without first determining the general solution.

② Non-Homogeneous eqns are solved without obtaining the Complementary integral.

③ L.T. is applicable not only to continuous funns but also to piecewise continuous funns, periodic funns, step funns ...

Applications:- L.T. is very useful in obtaining solns of L.D.E, both ODE & PDE, simultaneous DEs, I.Es, ...

Definition:- Let $f(t)$ be a function of 't' defined for all positive values of 't'. Then Laplace transforms of $f(t)$ is denoted by "$L\{f(t)\}$" is defined by

$$\boxed{L\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt = \bar{f}(s) \longrightarrow (1)}$$

provided that the integral exists. Here the parameter 's' is a real (or) complex number.

$$(1) \Rightarrow \quad f(t) = L^{-1}\{\bar{f}(s)\}$$

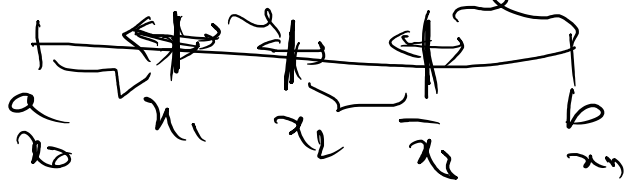
In such a case the function $f(t)$ is called the Inverse Laplace transform of $\bar{f}(s)$.

Conditions for L.T.:-

Exponential order:- A function $f(t)$ is said to be of exponential order 'a' If $\lim_{t \to \infty} e^{-st} f(t) = a$. finite quantity

Ex:- ① $t^2 \rightarrow$ ✓ ② $e^{t^3} \rightarrow e^{t^3}$ ✗

Piece-wise Continuous funn:- A funn $f(t)$ is said to be piece-wise contn over the closed interval $[a, b]$ if it is defined on that interval and is such that the interval can be divided into a finite no. of sub intervals, in each of which $f(t)$ is continuous & has both right and Left hand limits at every end point of the sub interval.



Sufficient conditions for the existence of the L.T. of a funn:-

The funn $f(t)$ must satisfy the following conditions for the existence of the L.T.

(i) The funn $f(t)$ must be piece-wise continuous (or sectionally continuous) in any limited interval $0 < a \le t \le b$.

(ii) The funn $f(t)$ is of exponential order.

Laplace Transforms of standard funns:-

(1) $L\{1\} = \int_0^\infty e^{-st} \cdot 1 \, dt = \left. \frac{e^{-st}}{-s} \right|_0^\infty = \frac{-1}{s}\left[e^{-\infty} - e^{0}\right]$

$= \frac{-1}{s}\left[\frac{1}{e^{\infty}} - 1\right] = \frac{1}{s}$

Sol:-

$$\boxed{L\{1\} = \frac{1}{s}}$$

② $L\{t\} \Rightarrow f(t) = t.$

Sol: $L\{t\} = \int_0^\infty e^{-st} \cdot t\,dt$

$$= t \cdot \frac{e^{-st}}{-s} - \int 1 \cdot \frac{e^{-st}}{-s}\,dt \Big|_0^\infty$$

$$= \underbrace{t \cdot \frac{e^{-st}}{-s}}_{\downarrow\, 0} + \frac{1}{s} \int \frac{e^{-st}}{-s} \Big|_0^\infty$$

$$= \frac{-1}{s^2}\left[e^{-\infty} - 0\right] = \frac{1}{s^2}$$

$$\boxed{L\{t\} = \frac{1}{s^2}}$$

③ $L\{t^n\}$

Sol: $\Rightarrow \int_0^\infty e^{-st} \cdot t^n\,dt.$

$$\underbrace{t^n \frac{e^{-st}}{-s}\Big|_0^\infty}_{\downarrow\, 0} - \int_0^\infty n \cdot t^{n-1} \frac{e^{-st}}{-s}\,dt$$

$$\Rightarrow \frac{n}{s} \int_0^\infty t^{n-1} \cdot e^{-st}\,dt$$

$$L\{t^n\} = \frac{n}{s} L\{t^{n-1}\}$$

$$L\{t^{n-1}\} = \frac{n-1}{s} L\{t^{n-2}\}$$

$$\vdots$$

$$L\{t^n\} = \frac{n}{s} \cdot \frac{n-1}{s} \cdots \frac{1}{s} L\{t^{n-n}\}$$

$$\boxed{L\{t^n\} = \frac{n!}{s^{n+1}}}$$

$$L\{t^n\} = \frac{n!}{s^{n+1}}$$

$$L\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}}$$

③ $L\{e^{at}\} \rightarrow f(t) = e^{at}$

Sol: $L\{e^{at}\} = \int_0^\infty e^{-st} \cdot e^{at}\, dt$

$$= \int_0^\infty e^{-(s-a)t}\, dt$$

$$= \left. \frac{e^{-(s-a)t}}{-(s-a)} \right|_0^\infty$$

$$= \frac{-e^{-\infty}}{s-a} + \frac{e^{0}}{s-a}$$

$$= \frac{1}{s-a} \quad \text{if } s > a$$

$$L\{e^{-at}\} = \frac{1}{s+a} \quad \text{if } s > -a.$$

$$\boxed{L\{e^{at}\} = \frac{1}{s-a}}$$

$$\boxed{L\{e^{-at}\} = \frac{1}{s+a}}$$

① $L\{\sinh at\}$.

Sol^n:- $L\{\sinh at\} = L\left\{\dfrac{e^{at} - e^{-at}}{2}\right\}$

$= \frac{1}{2} L\{e^{at}\} - \frac{1}{2} L\{e^{-at}\}$

$= \frac{1}{2} \frac{1}{s-a} - \frac{1}{2} \frac{1}{s+a}$

$= \frac{1}{2}\left\{\dfrac{s+a-s+a}{s^2-a^2}\right\} = \dfrac{a}{s^2-a^2}$

$$\boxed{L\{\sinh at\} = \frac{a}{s^2-a^2}}$$

② $L\{\cosh at\}$

Sol^n: $\cosh at = \dfrac{e^{at} + e^{-at}}{2}$

$L\{\cosh at\} = \frac{1}{2}\left\{\dfrac{s+a+s-a}{s^2-a^2}\right\}$

$$\boxed{L\{\cosh at\} = \frac{s}{s^2-a^2}}$$

③ $L\{\sin at\} = \int_0^\infty e^{-st} \sin at\, dt$

Sol^n: $\int e^{ax} \sin bx\, dx =$

$= \dfrac{e^{ax}}{a^2+b^2}\{a \sin bx - b\cos bx\}$

$$\boxed{L\{\sin at\} = \frac{a}{s^2+a^2}}$$

④ $L\{e^{iat}\} = L\{\overset{\cos at}{\cos at}\} + iL\{\overset{\sin at}{\sin at}\}$

Sol:- $L\{e^{iat}\} = \dfrac{1}{s-ia} \times \dfrac{s+ia}{s+ia}$

$$= \frac{s}{s^2+a^2} + i \cdot \frac{a}{s^2+a^2}$$

$$\boxed{L\{\sin at\} = \frac{a}{s^2+a^2}}$$

$$\boxed{L\{\cos at\} = \frac{s}{s^2+a^2}}$$

① $L\{(t^2+1)^2\}$

Sol:- $(t^2+1)^2 = t^4 + 2 \cdot t^2 + 1$

$$L\{t^n\} = \frac{n!}{s^{n+1}}$$

$$L\{(t^2+1)^2\} = L\{t^4\} + 2L\{t^2\} + L\{1\}$$

$$= \frac{4!}{s^5} + 2\frac{2!}{s^3} + \frac{1}{s}.$$

$$= \frac{1}{s^5}(24 + 4s^2 + s^4)$$

② $L\left\{\frac{e^{-at}-1}{a}\right\}$

Sol:- $\Rightarrow \frac{1}{a} L\{e^{-at}\} - \frac{1}{a} L\{1\}$

$\Rightarrow \frac{1}{a} \frac{1}{s+a} - \frac{1}{a} \cdot \frac{1}{s}$

$\Rightarrow \frac{1}{a}\left[\frac{\cancel{s}-\cancel{s}-a}{s(s+a)}\right]$

$\Rightarrow \frac{1}{\cancel{a}}\left(\frac{-\cancel{a}}{s(s+a)}\right)$

$= \frac{-1}{s(s+a)}$.

③ $L\{\sin 2t \cdot \cos t\}$

Sol:- $\sin 2t \cdot \cos t = \frac{1}{2} \cdot [2 \sin 2t \cos t]$

$= \frac{1}{2}[\sin 3t + \sin t]$

$L\{\sin 2t \cdot \cos t\}$

$= \frac{1}{2} L\{\sin 3t\} + \frac{1}{2} L\{\sin t\}$

$= \frac{1}{2} \frac{3}{s^2+9} + \frac{1}{2} \frac{1}{s^2+1}$.

Simplifying

H.W. ① $\cosh^2 2t$ ② $\cos^3 3t$

③ $\sin^2 t$ ④ $(\sin t + \cos t)^2$

① $L\{\sqrt{t}\}$

Sol: $L\{t^{1/2}\} =$

$$L\{t^n\} = \frac{n!}{s^{n+1}} = \frac{\Gamma(n+1)}{s^{n+1}}$$

$n \geq 1$ Natural

$0 < n < 1$

$\frac{a}{b}$, $b \neq 0$

$$L\{t^{1/2}\} = \frac{\Gamma(1/2+1)}{s^{1/2+1}}$$

$$= \frac{\frac{1}{2}\Gamma(1/2)}{s^{3/2}}$$

$$L\{t^{1/2}\} = \frac{\sqrt{\pi}}{2\, s^{3/2}}$$

H.W.

① $L\{\sin(\omega t + \alpha)\}$

$\omega, \alpha \to$ Constants.